

SHM

$F = kx$ Restoring $F_R = -kx$ $F = mg = ma$

$$\frac{d\vec{p}}{dt} = \vec{F} = -k\vec{x}$$

$$p = mv$$

$$\frac{d^2x}{dt^2} = \omega^2 x = a$$

$$\omega = 2\pi f = \frac{2\pi}{T}$$

$$\omega^2 = \frac{k}{m} \rightarrow \text{rad s}^{-1}$$

$x = A \sin(\omega t + \phi)$ $\rightarrow$ Displacement equation
 $x = A \cos(\omega t + \phi)$ $\rightarrow$ equilibrium
 $\rightarrow$ pulled/pushed

$$v = \frac{dx}{dt} = -A\omega \sin(\omega t + \phi) \rightarrow \text{Velocity}$$

$$v_{\max} = -A\omega$$

$$a = \frac{dv}{dt} = -A\omega^2 \cos(\omega t + \phi) \rightarrow \text{acceleration}$$

$$a_{\max} = -A\omega^2$$

$$E_p = \frac{1}{2} kx^2$$

$$E_p = \frac{1}{2} kA^2 \cos^2 \omega t \rightarrow \text{potential energy}$$

$$E_k = \frac{1}{2} mv^2$$

$$E_k = \frac{1}{2} kA^2 \sin^2 \omega t \rightarrow \text{kinetic energy}$$

$$E_T = E_p + E_k = \frac{1}{2} kA^2 \rightarrow \text{Total energy}$$

$$v = \omega \sqrt{A^2 - x^2}$$

$$a = -\omega^2 x$$

$\rightarrow$ No. Need time

$$1 \text{ N/m} = 1000 \text{ dyn/cm}$$

Pendulum

Tension $T = mg \cos \theta$

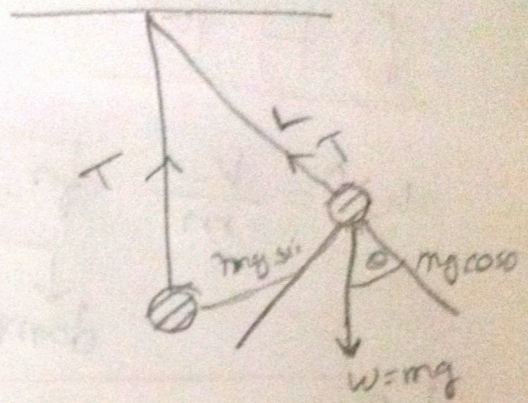
Restoring $F = -mg \sin \theta$

$$x = L\theta \quad \omega^2 = \frac{g}{L}$$

x = displacement (linear)

θ = angular displacement

$$\frac{d^2\theta}{dt^2} + \omega^2\theta = 0 \rightarrow \text{angular displacement.}$$



LC - Circuit / Resonant

$$V_L = L \frac{di}{dt}$$

Inductor

$$V_C = \frac{Q}{C}$$

Capacitor

$$i = \frac{dQ}{dt}$$

$$\omega_0^2 = \frac{1}{LC}$$

$$Q = Q_0 \cos \omega t$$

$$I = -I_0 \sin \omega t$$

$$\frac{2\pi}{T} = \frac{1}{\sqrt{LC}}$$

$$T = 2\pi\sqrt{LC}$$

$$f = \frac{1}{2\pi\sqrt{LC}}$$

Resonant frequency

D H M

$$\omega_n = \frac{k}{m}$$

$$r = \frac{b}{m}$$

damping constant

$$\text{Life-time} = \frac{1}{\gamma}$$

$$\frac{d^2x}{dt^2} + r \frac{dx}{dt} + \omega_n^2 x = 0 \rightarrow \text{eq of Damped H. M}$$

$$p = -\frac{\gamma}{2} \pm \sqrt{\frac{\gamma^2}{4} - \omega_n^2}$$

$$\omega_n^2 > \frac{\gamma^2}{4} \rightarrow \text{oscillatory} \rightarrow x = A e^{-\frac{\gamma t}{2}} \cos(\omega_1 t + \phi)$$

$$\omega_n^2 = \frac{\gamma^2}{4} \rightarrow \text{Critical}$$

$$\omega_n^2 < \frac{\gamma^2}{4} \rightarrow \text{over damped}$$

$$\omega_1 = \sqrt{\frac{k}{m} - \frac{\gamma^2}{4}}$$

$$T = \frac{2\pi}{\omega_1}$$

$$f = \frac{\omega_1}{2\pi}$$

$$\text{Damped Energy } E(t) = \frac{1}{2} k A^2 e^{-\frac{\gamma t}{m}}$$

RLC

$$V_L = L \frac{di}{dt}$$

$$V_C = \frac{Q}{C}$$

$$\gamma = \frac{R}{L}$$

$$\omega_0^2 = \frac{1}{LC}$$

$$\left\{ \begin{array}{l} \frac{1}{LC} > \frac{R^2}{4L^2} \rightarrow \text{oscillatory} \rightarrow Q = A e^{-\frac{R}{2L}t} \cos(\omega_1 t + \phi) \end{array} \right.$$

$$\left\{ \begin{array}{l} \frac{1}{LC} = \frac{R^2}{4L^2} \rightarrow \text{Critical} \end{array} \right.$$

$$\left\{ \begin{array}{l} \frac{1}{LC} < \frac{R^2}{4L^2} \rightarrow \text{over} \end{array} \right.$$

$$\omega_1 = \sqrt{\frac{1}{LC} - \frac{R^2}{4L^2}}$$

$$\phi = \frac{\omega_1}{2\pi}$$

$$T = \frac{2\pi}{\omega_1}$$